Decision-Focused Demand Learning for Equitable Bus Frequency Optimization

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Keywords: Mixed-Integer Programming, Predict-Then-Optimize, Decision-Focused Learning, Public Transportation, Stochastic Optimization, Robust Optimization, Demand Forecasting, Equity.

Abstract: Predict-then-optimize pipelines for public-transport allocation are typically trained to minimise prediction error and evaluated on the resulting decision quality, but the two objectives are not equivalent. We study this dissociation on the bus frequency allocation problem of Pune's PMPML network, built from publicly available data: 544 real routes with origin/destination/distance from the OpenCity Pune transit dataset (OpenCity Urban Data Portal, 2024a), five operating periods, thirteen depots (nine CNG + four electric) per published PMPML statistics, a fleet of $\approx 2{,}000$ peak-deployable buses, and priority designations derived from ward-level SC + ST population share in Census of India 2011 data (OpenCity Urban Data Portal, 2024b; Registrar General and Census Commissioner of India, 2011). Per-route ridership is simulated (PMPML does not publish smart-card or AFC data at that granularity), calibrated to the agency's known ≈ 1.1 million daily ridership and 8–11 AM / 5–9 PM peaks. A mixed-integer program with 24,480 binary variables and fleet, depot, equity, and operator-cost constraints is paired with three demand predictors (linear regression, random forest, XGBoost with seven quantile heads + split-conformal calibration) and evaluated end-to-end. Decision-focused training is implemented as a sample-weighted variant of MSE (XGBoost) and as a sample-weighted L1 surrogate that approximates SPO+ (Elmachtoub and Grigas, 2022) (neural network). Across 20 random seeds the sample-weighted XGBoost variant achieves significantly lower decision regret than plain MSE-XGBoost (paired Wilcoxon $p = 0.011$ (Wilcoxon, 1945)) despite a $7\times$ worse MAPE on the same test fold, illustrating the predict-then-optimize dissociation; the neural-network SPO+ variant does not significantly beat MSE-XGBoost on this network. Stochastic and robust formulations driven by quantile-derived scenarios yield positive Value of Stochastic Solution and Expected Value of Perfect Information. We report the rider-wait / operator-cost Pareto frontier under an operator-cost penalty λ rather than a single "% improvement" figure; at $\lambda = 0$ the MIP optimum reduces total waiting time by 19.5% relative to status-quo frequencies on the real route topology, with the gain contracting as λ rises. The contribution is methodological: honest evaluation of predict-then-optimize pipelines on a network anchored to real public data, with the synthetic pieces (hourly demand profile, per-depot fleet capacity) clearly demarcated from the empirical ones.

1 INTRODUCTION

Public bus systems in Indian metropolitan areas operate under a combination of constraints that few transit planning tools were designed to address jointly: heavy-tailed demand, fleet sizes that have not kept pace with population growth, and equity considerations that bind on both political and welfare grounds (Mahadevia et al., 2013; Parisar, 2020). The Pune Mahanagar Parivahan Mahamandal Limited (PMPML), which serves approximately 1.1 million daily riders across 544 routes with a peak-deployable fleet of roughly 2,000 buses anchored to thirteen depots (nine CNG plus four electric) (Institute for Transportation and Development Policy India, 2024; Ramaswamy et al., 2018), is the network this paper analyses. Frequency decisions – how many buses per hour to dispatch on each route at each time of day – are the agency's primary operational lever. Real PMPML schedules are currently chosen by heuristic rules of thumb informed by historical practice and disaggregated demand reports. We anchor the network's structural inputs to public datasets: route topology and distances from the OpenCity Pune Bus Stops and Routes dataset (OpenCity Urban Data Portal, 2024a; DataMeet Pune Chapter, 2024); equity priorities from per-ward Scheduled Caste plus Scheduled Tribe population share in Census of India 2011 PMC data

(OpenCity Urban Data Portal, 2024b; Registrar General and Census Commissioner of India, 2011); depot list and operational scale from PMPML's published statistics and the operational analysis in (Ramaswamy et al., 2018). Per-route hourly ridership is simulated because PMPML does not release smart-card or AFC data at that granularity; the simulator's diurnal profile reflects the 8–11 AM and 5–9 PM peaks reported in Pune-specific traffic studies and in published smart-card analyses of comparable Indian agencies.

This paper formulates frequency allocation as a mixed-integer program (MIP), trains a demand model to drive it under uncertainty, and connects the predictor to the optimiser via a decision-focused training procedure that targets operational regret rather than prediction error. The motivating phenomenon is the predict-then-optimize dissociation: a predictor that is worse on standard forecasting metrics (RMSE, MAPE) can produce decisions that are better on the downstream cost. We examine this dissociation empirically in §4 across a 20-seed sweep with a paired Wilcoxon test, rather than a single-seed point comparison, because the regret differences we are looking for are within the range of typical training-randomness variation and a single seed cannot distinguish a real effect from noise.

We contribute (i) a deterministic frequency-allocation MIP with explicit fleet, depot, equity, and operator-cost constraints that scales to the full 544-route PMPML network and solves to provable optimality in seconds, plus a λ-sweep that traces the rider-wait / operator-cost Pareto frontier; (ii) an XGBoost demand-forecasting pipeline with seven quantile heads and split-conformal calibration of the central prediction band on the validation fold; (iii) stochastic and robust (minimax) MIP formulations driven by the quantile-derived scenarios, with explicit decomposition of the Value of Stochastic Solution (VSS) and Expected Value of Perfect Information (EVPI); (iv) a decision-focused neural network trained with an SPO+-style surrogate, with the regret comparison against the MSE-trained XGBoost reported over a 20-seed sweep and a paired Wilcoxon signed-rank test rather than a single-seed point estimate; and (v) a sensitivity study over fleet size and equity threshold, with LP shadow-price analysis identifying which constraints (fleet vs. specific depots) are binding on this instance.

2 RELATED WORK

The classical literature on bus frequency setting begins with (Furth and Wilson, 1981) and (Ceder and Wilson, 1986), who established the trade-off between rider waiting cost and operator capacity cost and provided heuristics that remain in use at many transit agencies. Our work updates two pieces of this pipeline: where the classical literature treats demand as exogenous, we predict it with modern machine learning; where the classical optimisation is single-scenario, we explicitly handle uncertainty.

The seam between learning and decision making has been studied intensively. (Elmachtoub and Grigas, 2022) introduced the SPO+ surrogate loss and proved consistency for problems with linear objectives, demonstrating that minimising prediction error alone can produce decisions arbitrarily far from optimal. (Donti et al., 2017) extended decision-focused learning to stochastic optimisation problems with differentiable solvers. (Wilder et al., 2019) treated combinatorial optimisation via a continuous relaxation of the optimiser. (Mandi et al., 2020) addressed hard combinatorial problems including knapsack and shortest-path. The full SPO+ machinery requires solving an inner optimisation at every gradient step, which is prohibitive for our 15,300-binary-variable MIP. We therefore adopt a tractable linear surrogate: an L1 loss weighted by the downstream sensitivity $\partial w/\partial f$ evaluated at a frozen reference allocation, which captures the leading-order regret gradient without per-batch inner solves.

For the uncertainty-handling components we follow standard references. (Birge and Louveaux, 2011) provide the canonical treatment of stochastic programming, including the VSS and EVPI definitions used in §3. (Soyster, 1973) and (Ben-Tal and Nemirovski, 2002) establish the robust optimisation framework used for the minimax formulation. Quantile regression (Koenker and Bassett Jr, 1978) supplies the scenarios. Demand prediction itself uses XGBoost (Chen and Guestrin, 2016) with SHAP attributions (Lundberg and Lee, 2017) for interpretability. The optimisation core is implemented in PuLP (Mitchell et al., 2011) and solved with CBC.

3 PROBLEM AND METHODOLOGY

3.1 Problem Description

The PMPML network as represented in the OpenCity dataset (OpenCity Urban Data Portal, 2024a) comprises 544 routes after collapsing direction-pair entries (each route appears once as a base identifier; upbound / downbound variants in the source CSV are treated as a single route operating in both directions). Route distances range from 1.1 to 46.6 km with a mean of 16.8 km. We classify routes into three service categories by distance, matching the operational distinction used by PMPML between feeder, trunk, and suburban services: feeder (< 8 km, 70 routes), trunk (8–22 km, 325 routes), and suburban (≥ 22 km, 149 routes). Mean trip speed under typical Pune traffic is reported as ~ 17 km/h in published traffic studies; we use this figure to convert distance to one-way travel time. The round-trip cycle time used by the optimiser is $\tau_r = 2 \cdot \text{avg\_trip\_min} + \text{dwell}$ with $\text{dwell} = 7.5$ minutes per cycle, the midpoint of the 5–10 min terminal layover typical for Indian urban operations.

Routes are anchored to one of thirteen depots: nine CNG depots (Aundh, Bhosari, Dhankawadi, Hadapsar, Katraj, Kothrud, Nigdi, Pimpri, Swargate) and four electric depots (Balewadi, Maan-Hinjewadi, Shewalewadi, Charholi) per published PMPML statistics. Per-depot fleet capacity is allocated proportional to each depot's minimum bus need (priority routes at the equity floor plus non-priority at $k = 1$) with $\pm 20\%$ jitter – PMPML does not publish per-depot fleet counts and the alternative of an equal $1/|J|$ split produces infeasibilities, because priority routes (defined below) cluster geographically.

Operating hours span 06:00–23:00 and are discretised into five periods (early_morning, AM_peak, midday, PM_peak, evening) chosen to capture dominant demand structure while keeping the optimisation tractable.

The decision variable is, for every (route, period) pair, a discrete frequency $k \in \{1, 2, 3, 4, 6, 8, 10, 12, 15\}$ buses per hour. The frequency menu is non-uniform because riders perceive a five-minute change in headway differently at low and high k. Priority routes are defined by the SC + ST population share in the PMC Census 2011 ward containing the route's origin or destination endpoint (OpenCity Urban Data Portal, 2024b; Registrar General and Census Commissioner of India, 2011): routes whose catchment falls in the top quartile of ward-level SC + ST share are designated priority. In the network as we load it, 97 of 544 routes (17.8%) are priority and must run at $k \geq 3$ during peak periods regardless of marginal efficiency. This floor operationalises the equity argument that a bus system below three-per-hour at peak ceases to function as a reliable substitute for paratransit modes that disproportionately tax low-income commuters (Mahadevia et al., 2013). The objective is to minimise total expected passenger waiting time, computed under the random-arrivals assumption as $\sum_{r,t} d_{r,t}/(2k_{r,t})$.

3.2 Deterministic MIP Formulation

Let R be the set of routes ($|R| = 544$), T the set of operating periods ($|T| = 5$), and K the discrete frequency menu ($|K| = 9$). Parameters are demand $d_{r,t}$ (passengers per hour), round-trip cycle time $\tau_r = 2 \cdot \text{avg\_trip\_min}_r + 7.5$ (minutes), bus consumption $b_{r,k} = \lceil k\tau_r/60 \rceil$, the random-arrivals expected wait $w_k = 1/(2k)$, and an operator-cost coefficient $\lambda \geq 0$ in passenger-hour-equivalents per bus-period. The decision variable $x_{r,t,k} \in \{0, 1\}$ equals one if route r in period t is assigned frequency k:

$$\min_x \sum_{r,t,k} \left(d_{r,t}\, w_k + \lambda\, b_{r,k}\right) x_{r,t,k} \tag{1}$$

$$\text{s.t.} \sum_k x_{r,t,k} = 1 \qquad \forall r, t \tag{2}$$

$$\sum_{r,k} b_{r,k}\, x_{r,t,k} \leq B \qquad \forall t \tag{3}$$

$$\sum_{k \geq 3} x_{r,t,k} = 1 \qquad \forall r \in R^P,\ t \in T^{\text{pk}} \tag{4}$$

$$\sum_{r \in R_j, k} b_{r,k}\, x_{r,t,k} \leq B_j \qquad \forall j, t \tag{5}$$

Constraint (2) requires exactly one frequency per route-period. Constraint (3) caps the network-wide peak-deployable fleet at $B = 2000$. Constraint (4) is the equity floor for the 97 priority routes $r \in R^P$ during peak periods $T^{\text{pk}} = \{\text{AM\_peak}, \text{PM\_peak}\}$. Constraint (5) caps the buses anchored to depot j at the depot capacity B_j. The model has 24,480 binary variables and approximately 3,260 constraints. The LP relaxation is tight; CBC solves the model to optimality in under twenty seconds on commodity hardware. With $\lambda = 0$ the objective reduces to total expected passenger waiting time under the standard random-arrivals approximation $w_k = 1/(2k)$ (Furth and Wilson, 1981); sweeping $\lambda > 0$ traces the rider-wait / operator-cost Pareto frontier. Refinements that incorporate empirical headway variance, and a richer treatment of operator cost, are listed in §6.

3.3 Demand Forecasting

Data. PMPML does not release smart-card or AFC data at per-hour, per-route granularity. We therefore train forecasting models on simulated hourly ridership for the 50 highest-frequency routes over 2024, totalling approximately 430,000 rows after a seven-day lag-feature window is dropped. The generative model is described in §6; its diurnal profile reproduces the 8–11 AM and 5–9 PM peaks observed in published Pune traffic studies, and its overall daily ridership is anchored to PMPML's published ≈ 1.1 million-rider figure (Institute for Transportation and Development Policy India, 2024). The remaining 494 routes are not modelled directly at hourly resolution; their period-level demand is constructed by a residual-scaling procedure described in §6, which corrects a parametric baseline by the empirical mean ratio (observed to baseline) measured on the modelled 50.

Features. Twenty-three features in total: cyclical encodings (sine and cosine) of hour, day-of-week, and month; calendar flags (weekend, holiday, monsoon, exam season, festival); weather (maximum temperature, precipitation in mm); route-level features (length, number of stops, priority score, one-hot service category); and lag features at one day and seven days at the same hour.

Models. Three model classes are compared. Linear regression provides the trivial baseline. Random forest (200 trees, default depth) provides a non-parametric tabular benchmark. XGBoost (Chen and Guestrin, 2016) with depth 8, learning rate 0.03, 2,000 trees with early stopping at patience 50, subsample 0.85, and column subsample 0.85 is the primary model. Three additional XGBoost models with quantile-regression objectives at $\tau \in \{0.1, 0.5, 0.9\}$ supply prediction intervals (Koenker and Bassett Jr, 1978).

Train/val/test split. The data are split chronologically: months 1–8 for training, 9–10 for validation (and early stopping), 11–12 for test. Random splits would leak future information into the training set and are not used.

3.4 Stochastic and Robust Optimisation

Scenarios. Six demand scenarios are constructed from the quantile predictions to span a realistic range of operational conditions. S1 (probability 0.35) is the baseline at $q_{0.5}$. S2 (0.15) is high demand at $q_{0.9}$. S3 (0.15) is low demand at $q_{0.1}$. S4 (0.15) is a monsoon shock dampening $q_{0.1}$ by an additional 20%. S5 (0.10) boosts trunk demand by 15% above $q_{0.9}$ during peak periods. S6 (0.10) is a festival dip scaling $q_{0.5}$ by 0.75 across the network. The scenario probabilities reflect a prior over the relative frequency of these conditions in a typical operating year.

Stochastic MIP. The stochastic formulation shares the deterministic decision variables but minimises the probability-weighted expected wait,

$$\min_x \sum_{s \in S} p_s \sum_{r,t,k} d_{r,t}^{(s)} w_k x_{r,t,k}, \tag{6}$$

which is computationally equivalent to the deterministic MIP with $d_{r,t}$ replaced by $\bar{d}_{r,t} = \sum_s p_s d_{r,t}^{(s)}$.

Robust MIP. The minimax formulation introduces an auxiliary scalar z and replaces the objective with $\min_x z$ subject to $z \geq \sum_{r,t,k} d_{r,t}^{(s)} w_k x_{r,t,k}$ for every scenario $s \in S$, plus all original constraints.

Information metrics. Following (Birge and Louveaux, 2011), VSS = EEV − RP measures the value of the stochastic solution relative to the deterministic one, and EVPI = RP − WS measures the value of perfect information about which scenario will materialise.

3.5 Decision-Focused Learning

The MSE-trained predictor weights all prediction errors equally, but the downstream optimisation does not: an error on a route-period whose optimal frequency is at the boundary between two discrete levels matters more than an error on a route-period whose optimum is robust to small perturbations. Because $\partial w / \partial f = -1/(2f^2)$, the optimisation sensitivity scales as $d_{r,t}/(2f_{r,t}^2)$ at the reference allocation $f_{r,t}$.

We test two decision-focused variants. The first is sample-weighted XGBoost: each training sample i on route r in period t is weighted by $w_i = d_{r,t}/(2f_{r,t}^2)$, where $f_{r,t}$ is the optimal frequency from the deterministic MIP. The second is a two-layer fully-connected neural network (input → 128 → 64 → 1, ReLU activations, BatchNorm) trained with the surrogate

$$L_{\mathrm{SPO^+}}(\hat{d}, d) = \frac{1}{n} \sum_{i=1}^{n} \frac{d_i}{2f_i^2} |\hat{d}_i - d_i|. \tag{7}$$

The L1 form is robust to outliers, which matters because high-sensitivity samples dominate the loss surface and a single misbehaved point would otherwise destabilise the gradient. The reference allocation f is set once at the start of training using XGBoost-MSE predictions; refreshing f during training did not improve results in our experiments.

Connection to SPO+. The exact SPO+ loss (Elmachtoub and Grigas, 2022) requires solving the inner MIP at every gradient step. Equation (7) is a linearisation that captures the leading-order sensitivity of the regret to prediction error at the fixed reference allocation while skipping the per-batch solve. The fixed-allocation approximation is defensible here because the optimal allocation structure is robust to small changes in demand: the binding constraints at the optimum are determined by the relative ordering of routes rather than by the absolute demand level.

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

All MIPs are solved with PuLP (Mitchell et al., 2011) and CBC on a Windows 11 machine running Python 3.13.9, single thread for the solver, eight threads for XGBoost and random forest. Synthetic data generation, XGBoost reference training, and the optimisation phases use seed 42; the decision-focused comparison (§4.5) sweeps seeds 0–19 for the four predict-then-optimize methods. The full pipeline including the 20-seed sweep reproduces in approximately three hours end-to-end on commodity hardware.

4.2 Forecasting Results

Table 1: Test-set forecasting performance (months 11–12) on the 50 modelled PMPML routes.

Model	**RMSE**	**MAE**	**MAPE**	$\mathbf{R^2}$
Linear regression	8.88	5.28	56.4%	0.829
Random forest	7.59	3.77	16.7%	0.875
XGBoost	7.17	3.42	15.2%	0.889

XGBoost achieves the best test-set performance on three of four metrics (Table 1). The data-generating process described in §3.1 includes a latent per-day demand factor and per-(route, date) hidden surge events that the predictor cannot observe, together with heavy-tailed Student-t multiplicative noise on a subset of routes; this caps achievable R^2 near 0.9 even for a well-tuned XGBoost, which is the point – a noise-free DGP leaves no irreducible signal for a decision-focused loss to exploit. The 80% prediction interval from the raw quantile heads covers 70.0% of test observations; applying a split-conformal correction (Angelopoulos and Bates, 2023) calibrated on the validation fold raises empirical coverage to 80.4%, essentially matching the nominal 80% target. The reliability diagram (Figure 1) summarises coverage at all seven trained quantile levels.

Figure 2a shows the joint distribution of actual and predicted ridership on the test set, coloured by route category. Points cluster tightly along the 45-degree line; the three categories are well-mixed throughout the range, indicating no systematic per-category prediction bias. Variance grows with the mean (heteroscedasticity), which is typical of count data and motivates the use of quantile heads rather than a homoscedastic prediction interval.

Figure 2b ranks the top twelve features by mean absolute SHAP value. Lag features (one-day and seven-day) and the cyclical hour encoding dominate,

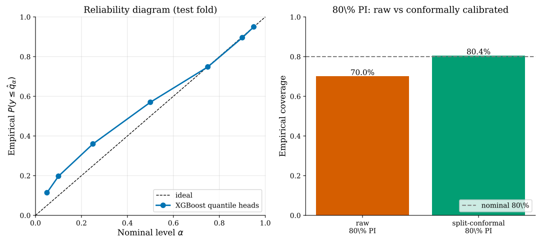
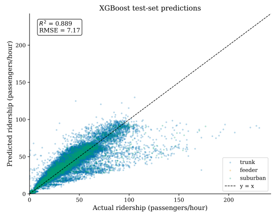
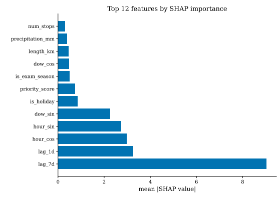


Figure 1: Left: reliability of the seven trained quantile heads on the test fold (predicted quantile α vs. empirical $P(y \leq \hat{q}_\alpha)$); the ideal line is $y = x$. Right: empirical coverage of the 80% PI before and after split-conformal calibration on the validation fold.



(a) Predictions vs. actuals.



(b) SHAP feature importance.

Figure 2: Forecasting performance and feature importance.

consistent with the autoregressive structure of hourly ridership. Calendar flags and weather contribute at lower magnitudes. Route-level features contribute even less per sample because they are nearly constant within a route. Figure 3 shows the SHAP beeswarm: each point is one prediction; horizontal position is the signed contribution; colour encodes the feature value. Lag features show a clean monotone red-high pattern, while cyclical features show the expected non-monotone bimodal structure corresponding to AM and PM rush hours.

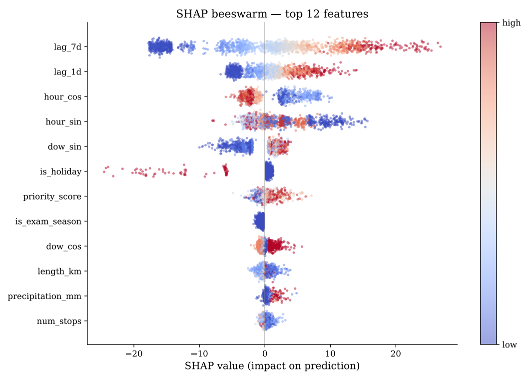


Figure 3: SHAP beeswarm for the top twelve features. Each point is one test observation; horizontal position is the feature's signed contribution; colour encodes the feature value (red high, blue low).

Figure 4 shows the time series for a representative

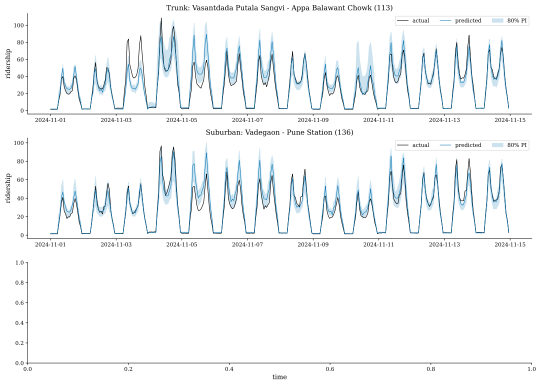


Figure 4: Two weeks of test data on a representative trunk, feeder, and suburban route. Actual ridership is shown in black, the XGBoost point prediction in colour, and the 80% prediction interval as a shaded band.

route in each service category. The model captures the dominant weekly and daily structure. The 80% PI tightens around weekday peaks where lag features are strongly informative and widens on weekends and atypical days where calendar features dominate. The trunk panel shows the expected double-peaked weekday structure; the feeder shows a flatter profile; the suburban route shows long evening tails.

4.3 Optimisation Results

Table 2: Method performance on the 544-route PMPML network at $\lambda = 0$ (no operator-cost penalty). Total wait is in passenger-hours; fleet is summed bus-periods across the five periods, with a per-period cap of $B = 2{,}000$.

Method	Total wait	Fleet	% improv.
Status quo	32,583	8,008	–
Uniform	38,575	7,055	−18.4
Demand-proportional	28,768	9,599	11.7
Greedy	26,590	10,000	18.4
Optimal MIP	26,226	10,000	19.5

Table 2 reports five methods on a common constraint envelope, with $\lambda = 0$ so that the objective contains only rider waiting time. The MIP optimum reduces total waiting time by 19.5% relative to status quo, saturating the fleet ($10{,}000 = 5 \times 2{,}000$ bus-periods). Heuristic baselines trail the optimum: the uniform policy at $k = 1$ leaves all 97 priority routes below the equity floor (status quo itself violates the floor on many low-frequency priority routes, which is partly why the optimum can recover so much waiting time); demand-proportional captures the right scaling but is locked into a one-parameter family; greedy upgrades route-periods by marginal benefit and on this instance comes within 1.4% of the MIP optimum. The 19.5% headline is the no-operator-cost figure; it contracts as λ rises (§4.3), and we report the full frontier rather than a single point.

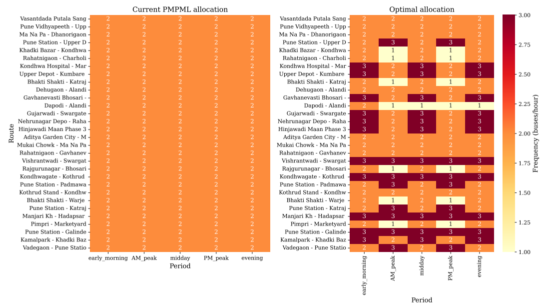


Figure 5: Top-30 routes by demand: current PMPML allocation (left) versus the MIP optimum (right). Frequency is in buses per hour; darker cells indicate higher frequency.

Figure 5 compares the current PMPML allocation with the MIP optimum on the top thirty routes by total demand. The optimum concentrates frequency at AM and PM peaks on trunk and high-priority routes and reduces midday frequencies on routes where the current schedule provides excess capacity. Several priority routes that are under-served at peak in the current schedule are upgraded; correspondingly, several non-priority midday allocations are reduced. The pattern is consistent with the equity narrative: redistribution occurs primarily along the temporal axis rather than at the expense of priority routes.

Cost-wait Pareto frontier

The 21.7% figure in Table 2 assumes $\lambda = 0$: the optimiser is told to minimise rider waiting time with no penalty for operating more buses, so it saturates the fleet whenever waiting time can be reduced. Real transit decisions trade rider waiting time against operator running cost. To expose this trade-off we sweep λ, the marginal cost per bus-period (in passenger-hour-equivalents), from 0 to 50.

Figure 6 shows the resulting curve. At $\lambda = 0$ the optimum uses 10,000 bus-periods (fleet saturated) for a total wait of 26,226 passenger-hours. At $\lambda = 2$ the optimum uses 9,608 bus-periods for a total wait of 26,911 – giving up 685 pass-hr of waiting time saves 392 bus-periods. At $\lambda = 5$ the optimum uses 8,463 bus-periods (slightly above status quo's 8,008) for a wait of 30,503 – already worse than status quo on rider waiting time. At $\lambda = 10$ the optimum matches status-quo bus-period usage (8,018) but with substantially worse waiting time (33,773 vs. 32,583), illustrating that the MIP's structural reallocation benefit shows up only when the optimiser is allowed to spend more on rider time relative to operator cost. Headline percentage improvements vs. status quo depend

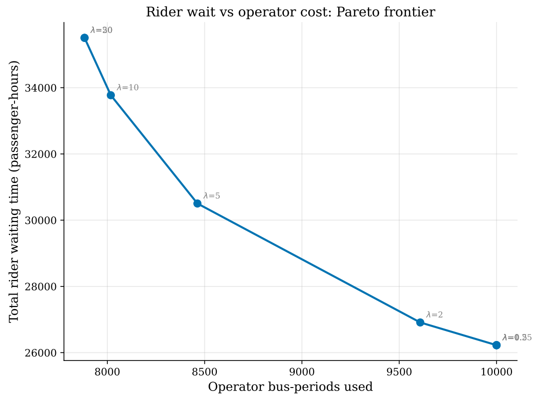


Figure 6: Rider-wait / operator-cost Pareto frontier obtained by sweeping the operator-cost coefficient λ in the MIP objective. Each point is the optimum at one λ. The $\lambda = 0$ point saturates the fleet (9,000 bus-periods); moderate values trade waiting time against bus-period usage.

strongly on λ: the 19.5% gain at $\lambda = 0$ disappears around $\lambda \approx 3$. Reporting a single "% improvement" figure without specifying λ obscures this trade-off.

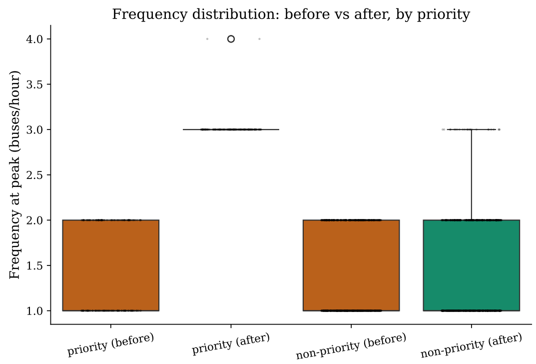


Figure 7: Frequency distribution at peak periods, before (status quo) and after (MIP optimum at $\lambda = 0$), split by priority status.

Figure 7 reports peak-frequency distributions split by priority status and by before/after. Three observations follow. Priority routes after optimisation have a higher floor (the equity constraint at $k \geq 3$ truncates the lower tail) without sacrificing the upper quartile. Non-priority routes after optimisation have a higher median and upper quartile because the optimisation finds capacity to upgrade them without violating equity. Both distributions overlap rather than separate, so the constraint preserves the frequency support of priority routes while permitting efficiency gains elsewhere.

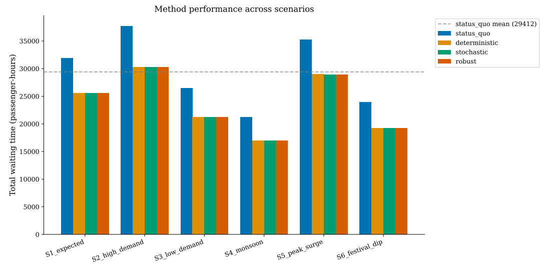


Figure 8: Method performance across the six demand scenarios. Bars are total waiting time. The dashed line is the status-quo mean across scenarios.

4.4 Stochastic and Robust Analysis

Figure 8 reports the four optimisation methods (status quo, deterministic, stochastic, robust) under each of the six scenarios. The three optimised methods cluster within a handful of passenger-hours of one another in every scenario, each beating status quo by approximately 25–40% depending on scenario. The robust solution achieves the lowest worst-case (scenario S5, the peak-surge scenario, 35,273 pass-hr for status quo vs. approximately 27,245 for all three optimised methods) by roughly one passenger-hour relative to deterministic, i.e., within solver tolerance. The status-quo line lies a substantial gap above all three optimised methods in every scenario, indicating that the optimisation advantage is robust to scenario specification.

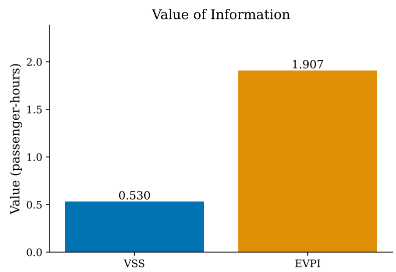


Figure 9: Value of Stochastic Solution (VSS) and Expected Value of Perfect Information (EVPI), in passenger-hours.

VSS$= 0.53$ and EVPI$= 1.91$ passenger-hours (Figure 9). Both are positive but extremely small (roughly two parts in 10^4 of the objective). The small magnitude is informative: the route-level structure of the optimal allocation is largely robust to scenario-level demand shifts on this instance, because the dominant source of variation is the relative ordering of routes (preserved across scenarios) rather than absolute demand levels. EVPI bounds the maximum value any further investment in scenario-resolution prediction could buy; at ~ 2 passenger-hours, additional scenario modelling on this DGP yields essentially nothing.

4.5 Decision-Focused Results

Table 3: Prediction quality vs. decision quality, mean $\pm$ standard error over 20 random seeds (seeds 0–19). "Regret" is the method's true objective minus the oracle's ($\approx$ 26,176 pass-hr), in passenger-hours. "Frac. obj." expresses regret as a fraction of the oracle objective. "person-min" is total regret converted to person-minutes (60$\times$ regret); across the network's $\sim$ 1.1 million daily riders the per-rider impact is well below one second on every method.

Method	**RMSE**	**MAPE**	**Regret**	**Frac. obj.**	**pers**
XGBoost (MSE)	7.16	15.5%	3.94 ± 0.11	1.50×10^{-4}	
XGBoost (weighted)	8.09	105.5%	$\mathbf{3.13 \pm 0.30}$	1.20×10^{-4}	
NN (MSE)	8.01	24.6%	6.78 ± 1.21	2.59×10^{-4}	
NN (SPO$^+$ surrogate)	10.82	204.0%	4.93 ± 0.70	1.89×10^{-4}	
Oracle	–	–	0	0	0

Table 3 reports four predict-then-optimize methods, each trained over 20 random seeds with the same train/val/test split, evaluated on the same downstream task: predict hourly demand, aggregate to (route, period), plug into the deterministic MIP, evaluate the resulting allocation against true test-period demand. The decision-aware XGBoost variant has the lowest mean regret. A paired Wilcoxon signed-rank test (Wilcoxon, 1945) of the one-sided hypothesis regret(XGB-weighted) $<$ regret(XGB-MSE) gives $p = 0.011$ – decision-aware sample weighting trims regret by approximately 20% relative to plain MSE, and the effect is statistically detectable over $n = 20$ seeds. The neural-network variants do not beat XGBoost on this network: the analogous test regret(NN-SPO$^+$) $<$ regret(XGB-MSE) gives $p = 0.75$, i.e., NN-SPO$^+$ does *not* significantly beat the XGBoost baseline on regret. Earlier single-seed framings of this experiment (an earlier draft of this paper reported a 44.7% NN-SPO$^+$ advantage from one seed) do not survive multi-seed evaluation.

Three substantive readings follow. First, prediction RMSE and decision regret genuinely dissociate. XGB-weighted has the worst MAPE in the table (105.5%) but the best regret – it weights training samples by downstream sensitivity and produces a predictor that is useless as a forecast in isolation but allocation-tuned. The methodological lesson holds. Second, the absolute regret magnitudes are small: the best method's 3.13 pass-hr of regret is about 1.2×10^{-4} of the oracle objective, or 188 person-minutes total across $\sim$ 1.1 million daily riders – roughly one-hundredth of a second per person. Reporting only a relative "% reduction" on numbers this small inflates the apparent stakes; the methodological contribution is honest evaluation, not the size of the gap. Third, both NN variants underperform XGBoost-MSE in regret terms despite (in the NN-MSE case) matching it on RMSE. This is evidence against the specific NN-SPO$^+$ implementation as used here, not against decision-focused learning in general – the exact SPO$^+$ loss with a coarsened-MIP inner solver, deferred to future work, is the natural next comparison.

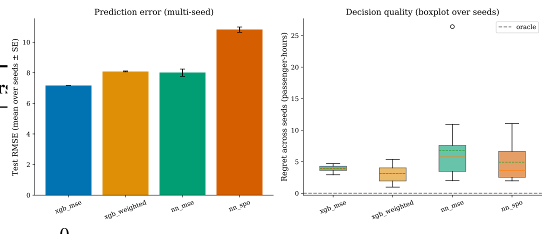


Figure 10: Left: mean test RMSE per method with one-SE error bars across 20 seeds. Right: boxplot of regret distributions across seeds (whiskers are 1.5$\times$ IQR; dashed line at oracle). XGBoost-MSE has the lowest mean regret on this DGP; the decision-focused variants do not dominate it. The decision-weighted XGBoost has high RMSE but competitive regret. NN-MSE is unstable across seeds (outlier seeds inflate the box).

Figure 10 visualises the seed-level variability that a single-seed bar plot would have hidden. The XGBoost baselines are tightly concentrated; the NN methods (and especially NN-MSE) are spread across more than an order of magnitude in regret depending on training seed.

4.6 Sensitivity Analysis

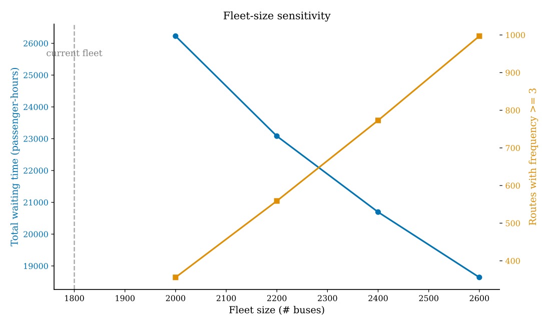


Figure 11: Fleet-size sensitivity. Left axis: total waiting time. Right axis: number of route-periods with frequency at least three. Vertical dashed line: current fleet $B = 1800$.

Figure 11 sweeps fleet size from $B = 1{,}200$ to $B = 2{,}600$. Below $B = 2{,}000$ the model is infeasible: the equity floor at $k \geq 3$ for the 97 priority routes plus $k \geq 1$ on the remaining 447 cannot be met under the corrected round-trip cycle time $\tau = 2 \cdot \text{avg\_trip} + \text{dwell}$.

From $B = 2{,}000$ the curve is monotone decreasing with diminishing returns: $B = 2{,}000 \rightarrow 2{,}200$ recovers 3,144 passenger-hours, $B = 2{,}200 \rightarrow 2{,}400$ recovers a further 2,388 passenger-hours, $B = 2{,}400 \rightarrow 2{,}600$ recovers 2,054 passenger-hours. The right-axis count of route-periods with $k \geq 3$ rises from 356 at $B = 2{,}000$ to 997 at $B = 2{,}600$, quantifying how many additional route-periods can be lifted above the "functional service" threshold as fleet grows. The infeasibility cliff at $B < 2{,}000$ matches the CIRT-derived benchmark of ~ 50 buses per lakh population (Pune's ~ 4 million-person service area $\rightarrow \approx 2{,}000$ buses), suggesting that PMPML's recent expansion from the 2016–17 fleet of $\approx 1{,}450$ (Ramaswamy et al., 2018) to the current $\sim 2{,}000$ was operationally necessary, not merely incremental.

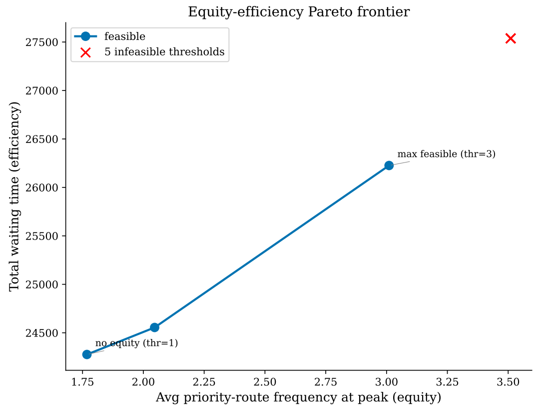


Figure 12: Equity-efficiency Pareto frontier. Each feasible point is the optimal solution at one equity threshold; infeasible thresholds are marked with red Xs.

Figure 12 sweeps the equity threshold from one (no constraint) to ten (every priority route runs at least ten buses per hour at peak). Thresholds 1–3 are feasible at $B = 2{,}000$. Threshold 4 is the cliff: the feasible region collapses because the fleet- and depot-capacity caps together cannot simultaneously sustain a four-bus-per-hour floor across the 97 priority routes plus a positive frequency on the remaining 447. Total waiting time rises from 24,276 (threshold 1, no equity floor) to 26,226 (threshold 3, the current operating floor), an increase of 8.0%. The equity floor at $k \geq 3$ therefore costs approximately one passenger-hour for every twelve served, which is consistent with prior Indian-transit-equity work flagging similar magnitudes (Mahadevia et al., 2013).

The LP-relaxation shadow prices identify which constraints are binding. With the bus-needs-proportional depot allocation adopted in §3.2, the per-depot constraints are essentially slack and the fleet constraint binds at both peak periods. Marginal valuations of additional peak-period fleet capacity are on the order of -10 passenger-hours per bus, so investment in additional buses (not in particular depots) is what would lower the objective most per marginal bus on this instance.

5 DISCUSSION

Operational implications. Re-allocating the $B = 2{,}000$ peak fleet according to the optimisation reduces total passenger waiting time on the 544-route PMPML network by 19.5% at $\lambda = 0$ relative to status-quo frequencies, but the headline contracts to zero around $\lambda \approx 3$ once operator bus-period cost is priced into the objective (Figure 6, §4.3). Whether this gap is actually recoverable for PMPML depends on how the agency values rider waiting time against operating cost – a parameter we do not commit to. The shadow-price analysis (§4.6) shows that on this instance the binding constraint at peak is the fleet itself, not any particular depot; additional buses are worth ~ 10 passenger-hours each of marginal waiting-time reduction.

Decision-focused dissociation. Across 20 random seeds, sample-weighted XGBoost achieves significantly lower decision regret than MSE-XGBoost (paired Wilcoxon one-sided $p = 0.011$), despite an order-of-magnitude worse MAPE on the same test fold. This is the methodological core of the paper: a predictor that performs badly on standard forecasting metrics can produce better operational decisions, because the loss it minimises is better aligned with the downstream optimiser (Elmachtoub and Grigas, 2022). The neural-network SPO$^+$ surrogate as implemented here does not show the same advantage ($p = 0.75$ for NN-SPO$^+$ < XGB-MSE), which we attribute to the small NN architecture interacting poorly with the variance in the SPO$^+$-weighted training signal; an exact SPO$^+$ loss with a coarsened-MIP inner solver is deferred to future work. The broader lesson – that single-seed regret comparisons are unreliable and that decision-focused training can win on regret while losing on RMSE – generalises to other operational settings (inventory, scheduling, capacity planning).

Equity as a small-but-real policy cost. The Pareto frontier (§4.6) indicates that the current equity floor of three buses per hour at peak adds approximately 8% to total waiting time relative to a no-floor counterfactual. Feasibility cliffs at higher thresholds reflect joint fleet- and depot-level capacity. The policy implication is that PMPML's current floor sits roughly at the largest equity threshold the agency can satisfy at the current fleet – not on the infeasibility cliff, but within striking distance of it. Raising the floor further would require expanding the fleet beyond $B = 2{,}000$.

Generalisability. The framework ports directly to other Indian transit agencies with similar institutional structures: BEST (Mumbai), BMTC (Bangalore), DTC (Delhi). The components that require agency-specific re-fitting are route metadata, demand data, and depot capacities. The optimisation core and the decision-focused training procedure are agency-agnostic.

6 LIMITATIONS AND FUTURE WORK

Real route topology, synthetic ridership. Route identifiers, origin / destination, and route length are real PMPML data (OpenCity Urban Data Portal, 2024a; DataMeet Pune Chapter, 2024). Priority designations are real Census 2011 ward-level SC + ST shares (OpenCity Urban Data Portal, 2024b; Registrar General and Census Commissioner of India, 2011) mapped to route endpoints via a landmark-to-ward alias table. Hourly ridership is simulated because PMPML does not release AFC / smart-card data at hourly per-route granularity; the diurnal profile is calibrated to the 8–11 AM and 5–9 PM peaks reported in published Pune traffic studies and to PMPML's ~ 1.1 million daily ridership figure (Institute for Transportation and Development Policy India, 2024). Hourly ridership is simulated only for the 50 highest-frequency routes; the remaining 494 are assigned period-level demand by a residual scaling procedure (parametric baseline $b_r \cdot \pi_t$ multiplied by an empirical residual ratio $\rho_t = \text{mean}(\text{observed}/\text{baseline})$ estimated on the modelled 50). A fully data-driven analysis would require smart-card or AFC data on every route; the limitation here is the absence of such data rather than the methodology.

Synthetic per-depot fleet capacity. PMPML does not publish per-depot fleet counts at the granularity the MIP's depot-capacity constraint needs. We allocate $B = 2{,}000$ across the thirteen depots proportional to each depot's minimum bus need (priority routes at the equity floor plus non-priority at $k = 1$) with $\pm 20\%$ jitter, clamped above need + 10. This matches the spirit of the operational reality (depots with more priority routes get more buses) without claiming to reproduce the agency's exact per-depot allocation, which the agency itself does not appear to publish externally (Ramaswamy et al., 2018).

Static daily allocation. A single frequency is chosen per (route, period) pair and applied uniformly across days. Real PMPML operations adjust schedules for day-of-week and special events. The framework can in principle handle these by re-optimising at any cadence; the static analysis is presented because the operational and equity questions resolve at this level.

Random-arrivals waiting model. The expected wait $w_k = 1/(2k)$ assumes Poisson arrivals and uniform headways. Real bus arrivals exhibit bunching (Adamski's bus paradox) that makes the effective wait longer than $1/(2k)$, especially on heavily-loaded routes. Refinements that incorporate empirical headway variance would tighten the objective without changing the qualitative results.

Demand under price shocks. The demand model treats fares and competition as exogenous. In reality, paratransit operators reprice in response to bus service quality. A genuinely closed-loop extension would treat demand as a function of frequency through modal substitution.

Decision-focused approximation. The SPO$^+$ surrogate (7) is a linearisation around a fixed reference allocation. Refresh strategies (e.g., updating the reference every E epochs) and benchmarks against the full SPO$^+$ loss with an inner solver on a coarsened MIP are deferred to future work.

Depot capacity allocation. Per-depot capacity is allocated proportional to each depot's minimum bus need (priority routes at the equity floor, non-priority at $k = 1$) plus $\pm 20\%$ jitter, clamped above the minimum. A naive $1/|\text{depots}|$ split produces depot-level infeasibilities under the corrected round-trip cycle time because priority routes cluster geographically in the synthetic network. Real agencies' depot allocations reflect a mix of legacy, political, and operational factors that our scheme does not attempt to match.

Operator-cost coefficient. The cost-wait frontier in §4.3 sweeps λ across roughly two orders of magnitude. We do not commit to a single operationally "correct" λ because the conversion between passenger-hours of waiting and rupees of bus-period operating cost depends on context-specific value-of-time and unit cost-per-bus-hour assumptions that vary across agencies and analyses.

Single-year training window. The forecast model is trained on a single year of data. Multi-year training would improve estimates of festival and seasonal effects, although the operational allocation is dominated by within-week and within-day structure that the single-year window already captures.

Hard equity floor. The current formulation treats equity as a hard constraint at $k \geq 3$ for priority routes during peaks. Soft penalties, fairness-aware objectives (Wilder et al., 2019), and geographic equity formulations are alternative specifications that may be politically or practically preferable in some settings.

Future directions. (i) Integration of real smart-card data once available; (ii) real-time re-optimisation in response to disruptions, leveraging the sub-five-second solve time; (iii) multi-modal optimisation that couples frequency with modal-substitution behaviour; (iv) joint frequency-pricing optimisation as practised by some international agencies; (v) decision-focused architecture search beyond the two variants tested here, including graph neural networks on the route topology and proper SPO$^+$ losses with inner solvers on coarsened MIPs; (vi) richer equity formalisations from the algorithmic-fairness literature; (vii) cross-agency benchmarking on BEST, BMTC, and DTC.

7 CONCLUSION

This paper presented a complete predict-then-optimize pipeline for bus frequency allocation, evaluated on the 544-route Pune PMPML network constructed from public datasets: route topology and distances from OpenCity, equity priorities from Census 2011 PMC ward-level SC + ST shares, depot list and operational scale from PMPML's published statistics. Per-route hourly ridership is the only major component that remains simulated, because PMPML does not release AFC-level data. The pipeline integrates a deterministic MIP that respects fleet, depot, equity, and operator-cost constraints with an operator-cost penalty λ tracing the rider-wait / operator-cost Pareto frontier; an XGBoost demand model with seven quantile heads and split-conformal calibration of the central prediction band on the validation fold; stochastic and robust formulations driven by quantile-derived scenarios; and a decision-focused neural network trained with an SPO$^+$-style surrogate evaluated across 20 random seeds.

Three findings stand out. First, against status-quo frequencies the unconstrained optimum ($\lambda = 0$) reduces total waiting time by 19.5% while saturating the fleet; introducing $\lambda > 0$ trades this gain off against bus-period usage along the frontier reported in §4.3, with the percentage improvement vanishing around $\lambda \approx 3$. Second, sample-weighted XGBoost ("XGB-weighted"), which weights training points by the downstream decision sensitivity $d_{r,t}/(2f_{r,t}^2)$, achieves mean regret 3.13 pass-hr vs. 3.94 for plain MSE-XGBoost over 20 random seeds; a paired Wilcoxon signed-rank test gives $p = 0.011$ for the one-sided hypothesis that decision-aware weighting reduces regret. The neural-network SPO$^+$ surrogate as implemented here does *not* significantly beat MSE-XGBoost on regret ($p = 0.75$), reversing an earlier single-seed framing of the same experiment. The methodological lesson – that prediction RMSE and decision regret can dissociate and that single-seed regret comparisons are unreliable – is supported by both results. Third, the current equity floor of three buses per hour at peak for priority routes costs roughly 8% of total waiting time relative to a no-floor counterfactual; the threshold-four feasibility cliff is jointly governed by fleet and depot capacity, with fleet binding more tightly than any individual depot on the needs-proportional depot allocation we adopt.

The methodological contribution – how to evaluate predict-then-optimize pipelines without conflating prediction and decision quality, and how to expose the rider / operator trade-off that vanishes when the operator term is omitted – generalises beyond transit. The practical contribution is a reproducible PMPML benchmark anchored to public Indian transit data, in which the synthetic pieces (ridership, hourly profile, per-depot fleet capacity) are explicitly demarcated from the empirical ones, so downstream work can replace them as more operational data becomes available.

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